

DSA Practical No: 04

Name: Daundkar Samarth Dattatray

PRN: B26ET2025

Div: SY3

Class: C3

INPUT :

```
#include <stdio.h>

#include <string.h>

#define MAX 30

// Employee structure
struct Employee
{

    int emp_no;
    char name[20];
    float salary;
};

void displayRecord(struct Employee emp)
{
    printf("\nEmployee No : %d", emp.emp_no);
    printf("\nName    : %s", emp.name);
    printf("\nSalary   : %f\n", emp.salary);
}

void create(struct Employee emp[], int *n)
{
    int i;
```

```

printf("\nEnter number of employees: ");
scanf("%d", n);

for (i = 0; i < *n; i++)
{
    printf("\nEnter details of employee %d\n", i + 1);

    printf("Enter employee number: ");
    scanf("%d", Cemp[i].emp_no);

    printf("Enter name: ");
    scanf("%19s", emp[i].name);

    printf("Enter salary: ");
    scanf("%f", Cemp[i].salary);
}

printf("\nEmployee database created successfully.\n");
}

// Display Function
void display(struct Employee emp[], int n)
{
    int i;

    if (n == 0)
    {

```

```

        printf("\nDatabase is empty.\n");
        return;
    }

    printf("\n===== EMPLOYEE DATABASE =====\n");

    for (i = 0; i < n; i++)
    {
        printf("\nEmployee %d", i + 1);
        displayRecord(emp[i]);
    }
}

// Search Function
void search(struct Employee emp[], int n)
{
    int i;
    int emp_no;
    int found = 0;

    printf("\nEnter employee number to search: ");
    scanf("%d", &emp_no);

    for (i = 0; i < n; i++)
    {
        if (emp[i].emp_no == emp_no)
        {
            printf("\nEmployee found!");

```

```

        displayRecord(emp[i]);

        found = 1;

        break;
    }
}

if (found == 0)
{
    printf("\nEmployee not found.\n");
}
}

// Modify Function
void modify(struct Employee emp[], int n)
{
    int i;
    int emp_no;
    int found = 0;

    printf("\nEnter employee number to modify: ");
    scanf("%d", &emp_no);

    for (i = 0; i < n; i++)
    {
        if (emp[i].emp_no == emp_no)
        {
            printf("\nEnter new name: ");
            scanf("%19s", emp[i].name);

```

```

printf("Enter new salary: ");
scanf("%f", Cemp[i].salary);

printf("\nEmployee record modified successfully.\n");

    found = 1;
    break;
}
}

if (found == 0)
{
    printf("\nEmployee not found.\n");
}
}

//Append Function
void append(struct Employee emp[], int *n)
{
    if (*n >= MAX)
    {
        printf("\nDatabase is full.\n");
        return;
    }

    printf("\nEnter employee number: ");
    scanf("%d", Cemp[*n].emp_no);

```

```

printf("Enter name: ");
scanf("%19s", emp[*n].name);

printf("Enter salary: ");
scanf("%f", Cemp[*n].salary);

(*n)++;

printf("\nEmployee added successfully.\n");
}

// Delete Function
void deleteEmployee(struct Employee emp[], int *n)
{
    int i, j;
    int emp_no;
    int found = 0;

    printf("\nEnter employee number to delete: ");
    scanf("%d", Cemp_no);

    for (i = 0; i < *n; i++)
    {
        if (emp[i].emp_no == emp_no)
        {
            for (j = i; j < *n - 1; j++)
            {

```

```

        emp[j] = emp[j + 1];
    }

    (*n)--;

    printf("\nEmployee deleted successfully.\n");

    found = 1;
    break;
}
}

if (found == 0)
{
    printf("\nEmployee not found.\n");
}
}

// Sorting
void sort(struct Employee emp[], int n)
{
    int i, j;
    struct Employee temp;

    for (i = 0; i < n - 1; i++)
    {
        for (j = 0; j < n - i - 1; j++)
        {

```

```

        if (emp[j].emp_no > emp[j + 1].emp_no)
        {
            temp = emp[j];
            emp[j] = emp[j + 1];
            emp[j + 1] = temp;
        }
    }
}

printf("\nEmployee database sorted by employee number.\n");
}

int main()
{
    struct Employee emp[MAX];

    int n = 0;
    int choice;
    char ch;

    do
    {

        printf("EMPLOYEE DATABASE\n");
        printf("1. Create Database\n");
        printf("2. Display Database\n");
        printf("3. Search Employee\n");
        printf("4. Modify Employee\n");
    }

```

```
printf("5. Append Employee\n");  
printf("6. Delete Employee\n");  
printf("7. Sort Database\n");  
printf("8. Exit\n");
```

```
printf("\nEnter your choice: ");  
scanf("%d", Cchoice);
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        create(emp, Cn);
```

```
        break;
```

```
    case 2:
```

```
        display(emp, n);
```

```
        break;
```

```
    case 3:
```

```
        search(emp, n);
```

```
        break;
```

```
    case 4:
```

```
        modify(emp, n);
```

```
        break;
```

```
    case 5:
```

```
        append(emp, Cn);
```

```
break;
```

```
case 6:
```

```
deleteEmployee(emp, Cn);
```

```
break;
```

```
case 7:
```

```
sort(emp, n);
```

```
break;
```

```
case 8:
```

```
printf("\nExit\n");
```

```
return 0;
```

```
}
```

```
if (choice != 8)
```

```
{
```

```
printf("\nDo you want to continue? (y/n): ");
```

```
scanf(" %c", Cch);
```

```
}
```

```
else
```

```
{
```

```
ch = 'n';
```

```
}
```

```
} while (ch == 'y' || ch == 'Y');
```

```
printf("\nProgram ended.\n");

return 0;

}
```

OUTPUT :

```
PS D:\dsa\Experiment-04-Employee-Database-Management> .\program.exe
EMPLOYEE DATABASE
1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit

Enter your choice: 1

Enter number of employees: 3

Enter details of employee 1
Enter employee number: 001
Enter name: sahil
Enter salary: 20000

Enter details of employee 2
Enter employee number: 002
Enter name: tushar
Enter salary: 20000

Enter details of employee 3
Enter employee number: 003
Enter name: samarth
Enter salary: 20000
```

Employee database created successfully.

Do you want to continue? (y/n): y

EMPLOYEE DATABASE

1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit

Enter your choice: 2

===== EMPLOYEE DATABASE =====

Employee 1

Employee No : 1

Name : sahil

Salary : 20000.000000

Employee 2

Employee No : 2

Name : tushar

Salary : 20000.000000

Employee 3

Employee No : 3

Name : samarth

Salary : 20000.000000

Do you want to continue? (y/n): y

EMPLOYEE DATABASE

1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit

Enter your choice: 3

Enter employee number to search: 002

Employee found!

Employee No : 2

Name : tushar

Salary : 20000.000000

Do you want to continue? (y/n): y

EMPLOYEE DATABASE

```
Do you want to continue? (y/n): y
EMPLOYEE DATABASE
1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit

Enter your choice: 4

Enter employee number to modify: 001

Enter new name: sahilmore
Enter new salary: 20000

Employee record modified successfully.

Do you want to continue? (y/n): y
EMPLOYEE DATABASE
1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
```

```
Enter your choice: 5

Enter employee number: 003
Enter name: samarth
Enter salary: 25000

Employee added successfully.

Do you want to continue? (y/n): y
EMPLOYEE DATABASE
1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit

Enter your choice: 6

Enter employee number to delete: 001

Employee deleted successfully.

Do you want to continue? (y/n): y
EMPLOYEE DATABASE
```

```
Do you want to continue? (y/n): y
EMPLOYEE DATABASE
1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit

Enter your choice: 7

Employee database sorted by employee number.

Do you want to continue? (y/n): y
EMPLOYEE DATABASE
1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit

Enter your choice: 2
```

```
===== EMPLOYEE DATABASE =====

Employee 1
Employee No : 2
Name       : tushar
Salary     : 20000.000000

Employee 2
Employee No : 3
Name       : samarth
Salary     : 20000.000000

Employee 3
Employee No : 3
Name       : samarth
Salary     : 25000.000000

Do you want to continue? (y/n): y
EMPLOYEE DATABASE
1. Create Database
2. Display Database
3. Search Employee
4. Modify Employee
5. Append Employee
6. Delete Employee
7. Sort Database
8. Exit
```